ETW3482 Data Mining and Predictive Modelling

Individual Project: Data Exploration and Data Preparation Project

Semester 2, 2026

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Report Word Count	1,890 words

Maximum 2,000 words. There is no additional percentage or word-count allowance.

1. Dataset Context and Overview

Lusitania Bank wants to rank existing customers by their likelihood of subscribing to a term deposit in an upcoming telephone campaign, so that limited call capacity is directed at the best prospects. Prediction occurs before contact. The binary target y records whether a customer subscribed: 11.3% said yes and 88.7% said no (Figure A3). The working dataset contains 8,000 observations and 15 potential predictors, 8 numerical and 7 categorical (Figure A6). They cover demographics (age, job, marital status, education), financial commitments (default, housing, personal loans), previous campaign contact, and macroeconomic indicators (employment variation, consumer prices and confidence, Euribor rates, employment levels).

2. Data Exploration and Critical Diagnosis

The Explore Raw node checked the untreated data. Issues are ranked by how many records they affect, how serious they are, and whether they can be fixed. Values that no screenshot shows were calculated from the CSV file (Table A1).

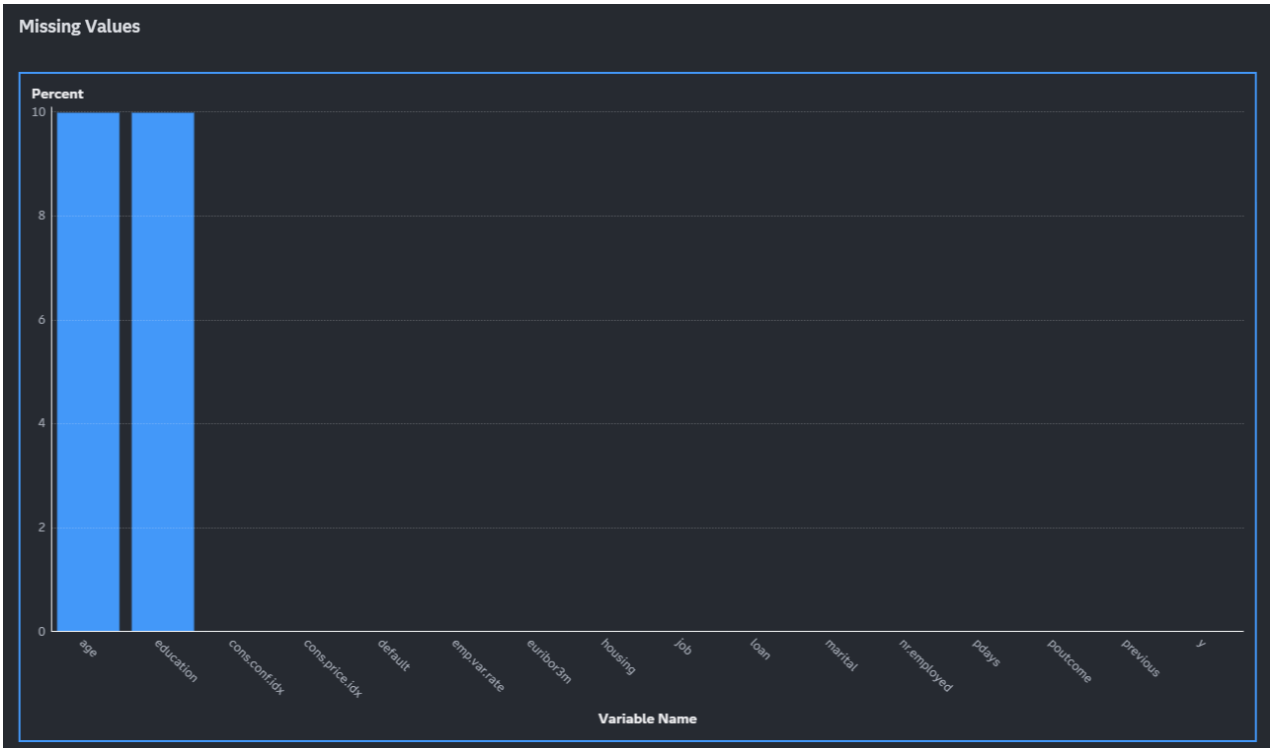


Figure 1: Initial missing values by variable.

About 10% of the values for age (numeric) and education (categorical) are missing, and no other variable has gaps (Figure 1). Only 91 customers are missing both, so 1,509 records, or 18.9% of the data, have at least one gap (Table A1). Many models skip incomplete records (Little & Rubin, 2019), so dropping them would lose all 1,509. These customers subscribe at about 11% to 12%, similar to everyone else (Table A1). That does not prove the data are missing completely at random (MCAR; Little & Rubin, 2019), but it gives no target-based reason to discard them. This is a high priority because it affects the largest share of records.

Interval Variable Moments

Variable Na...	Minimum	Maximum	Mean	Standard D...	Skewness	Kurtosis	Relative Va...	Mean plus 2...	Mean minus...
age	17	89	40.0600	10.4498	0.7755	0.7856	0.2609	60.9595	19.1605
cons.conf.idx	-50.8000	-26.9000	-40.5215	4.6052	0.2699	-0.3739	0.1136	-31.3111	-49.7319
cons.price.idx	92.2010	98.1261	93.6661	0.8012	2.1554	9.6095	0.0086	95.2686	92.0637
emp.var.rate	-3.4000	1.4000	0.0968	1.5581	-0.7404	-1.0323	16.0978	3.2129	-3.0193
euribor3m	0.6350	5.0450	3.6365	1.7275	-0.7303	-1.3749	0.4751	7.0915	0.1815
nr.employed	4,963.6000	5,228.1000	5,167.2073	72.1515	-1.0705	0.0679	0.0140	5,311.5102	5,022.9044
pdays	0	999	963.6258	184.0614	-5.0117	23.1237	0.1910	1,331.7485	595.5030
previous	0	7	0.1788	0.5151	3.9746	22.4130	2.8817	1.2090	-0.8515

Figure 2: Initial interval variable moments.

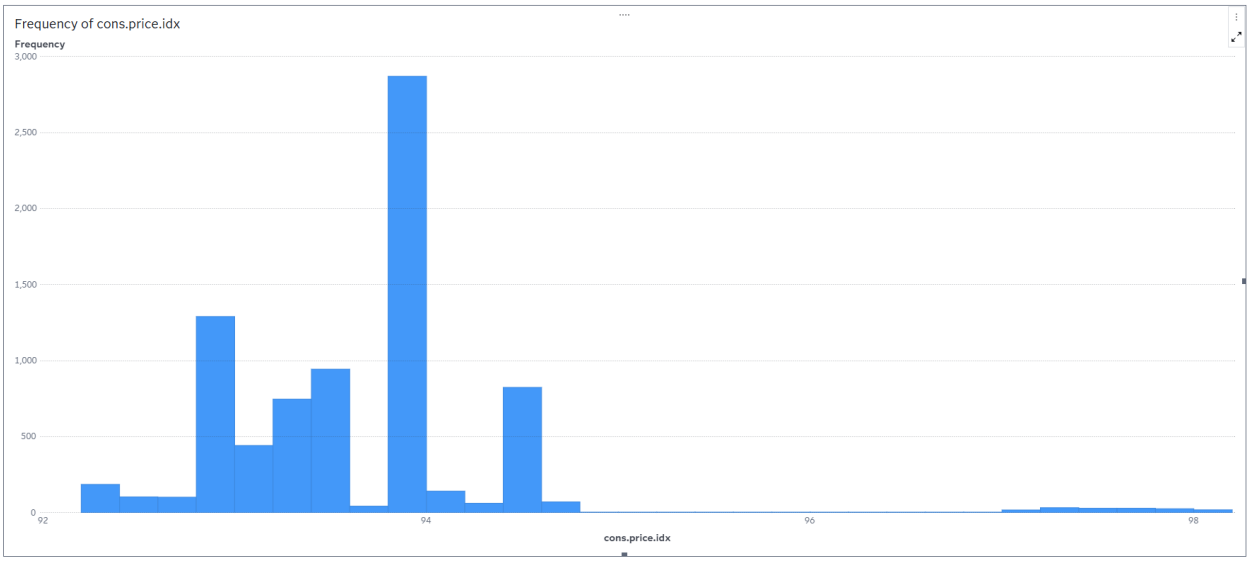


Figure 3: Distribution of cons.price.idx before treatment.

cons.price.idx has a skewness of 2.16 and a kurtosis of 9.61 (Figure 2), above commonly used reference values of 2 and 7 (Kim, 2013). The histogram shows a group of values between roughly 97 and 98.1, about 2.3 index points above the main body, which ends at 94.767 (Figure 3; Table A1). With quartiles of 93.075 and 93.994 (Figure A1), the IQR is 0.919 and the upper

fence is 93.994 + 1.5 × 0.919 = 95.3725 (Tukey, 1977). Exactly 160 observations, 2.0% of the data, lie above it, all above 97.0, while the minimum of 92.201 is above the lower fence (Figure A1; Table A1). This separate tail distorts the summary statistics and may overly influence models. Priority is high because the problem is serious and easy to separate.

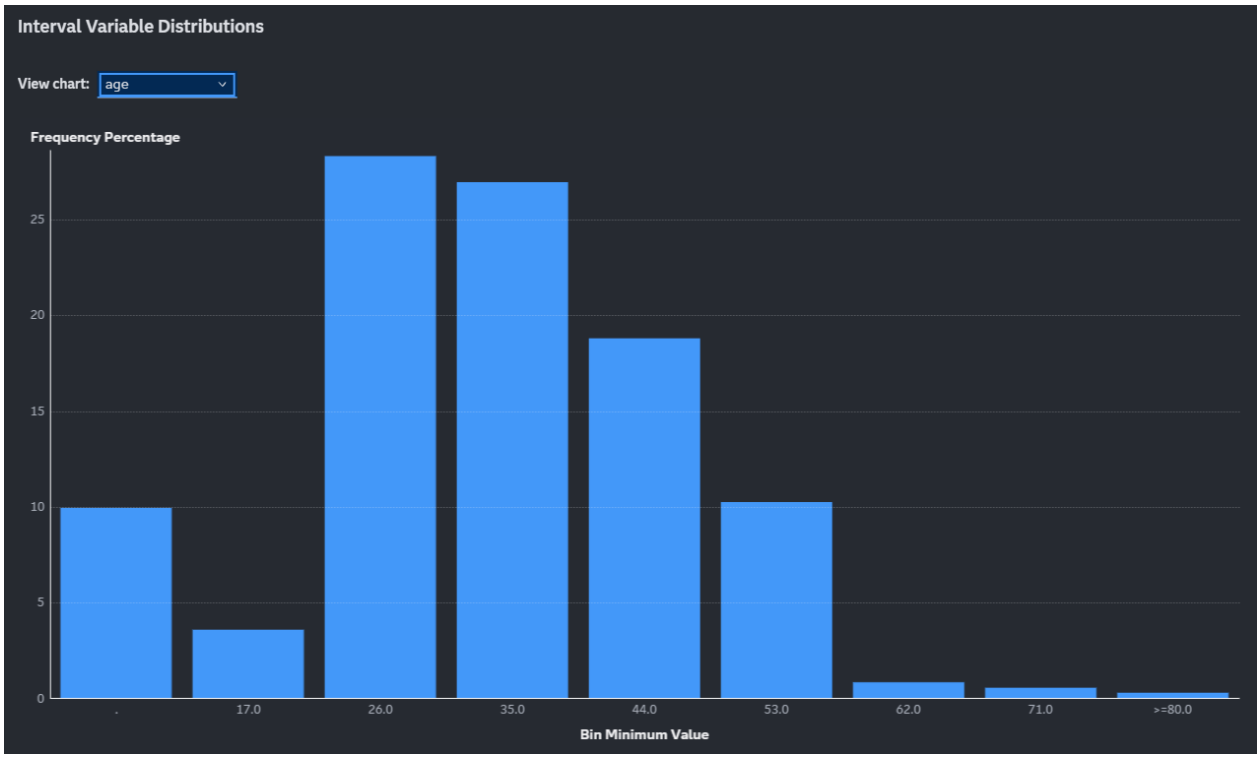


Figure 4: Distribution of age before treatment.

Age ranges from 17 to 89 and is mildly skewed towards older ages, with a skewness of 0.78 (Figure 2). The IQR rule would flag 84 customers aged 70 or over, using a fence of 69.5 (Table A1), but the histogram shows a gradual taper, not a separate group (Figure 4), and these ages are realistic for bank customers. Being rare does not mean being wrong, and capping or removing them would discard real information. No treatment is needed.

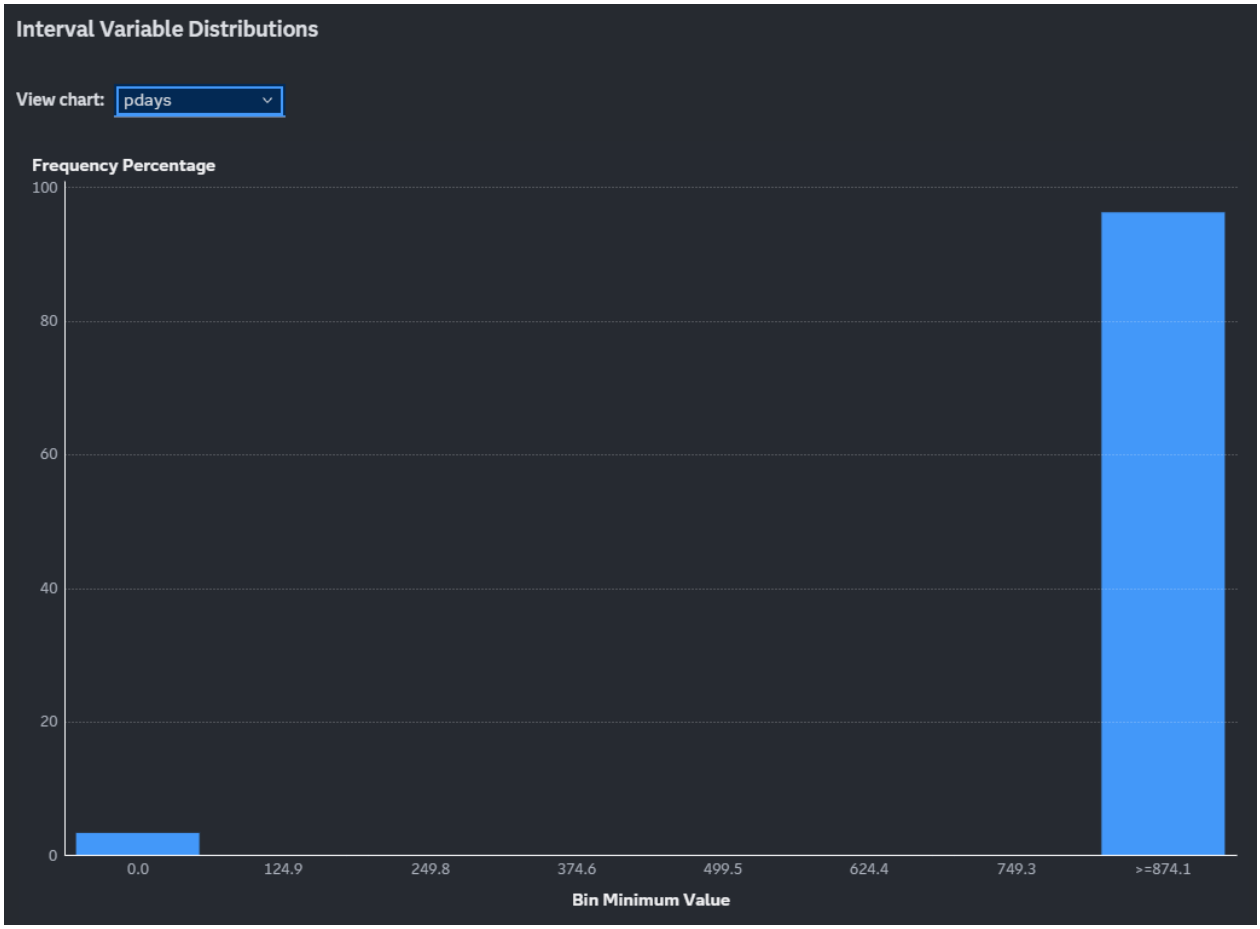


Figure 5a: Distribution of pdays.

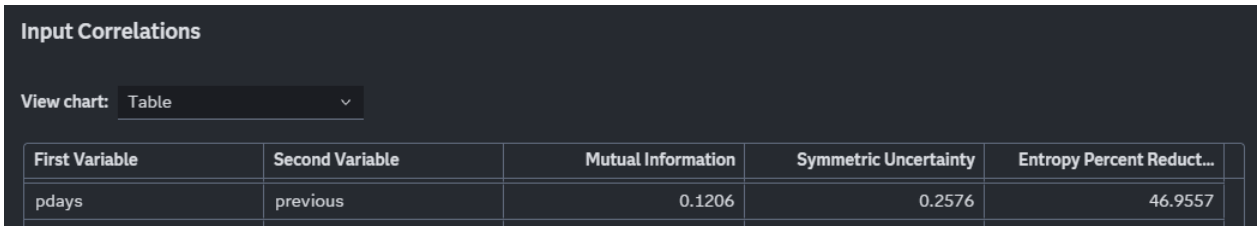


Figure 5b: Association between pdays and previous.

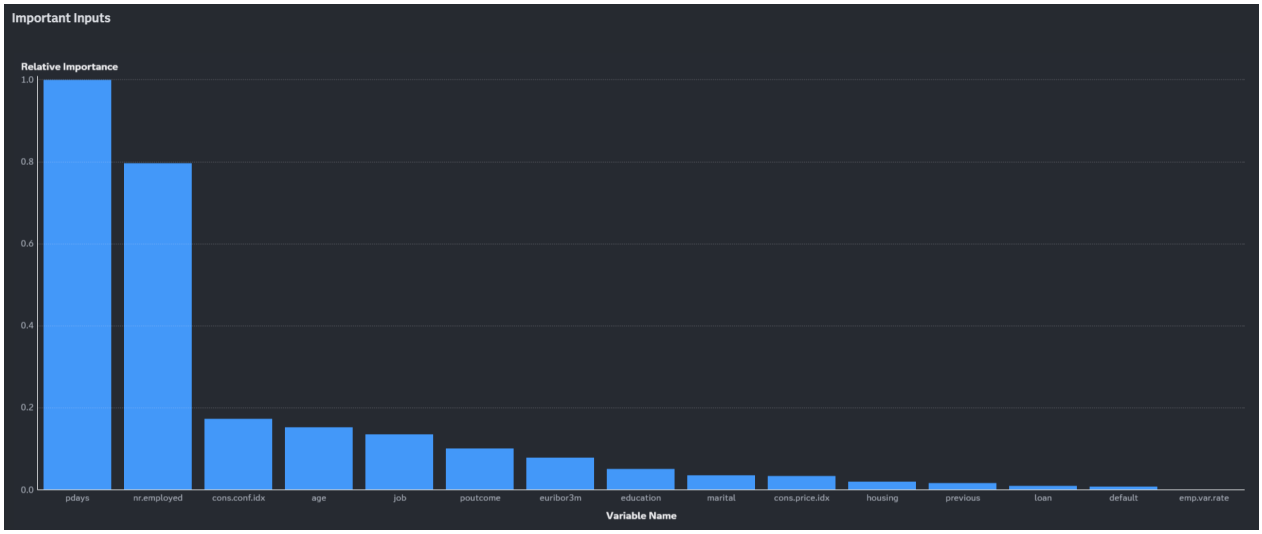


Figure 6a: Important Inputs chart from the Data Exploration node before any treatment.

Variable Importance			
	Training		
Variable	Importance	Relative Importance	Count
pdays	160.89	1.0000	2
nr.employed	128.47	0.7985	2
cons.conf.idx	28.2119	0.1754	3
age	24.7128	0.1536	12
job	21.9753	0.1366	10
poutcome	16.5593	0.1029	2
euribor3m	12.8230	0.0797	5
education	8.3538	0.0519	6
marital	5.8872	0.0366	5
cons.price.idx	5.6000	0.0348	5
housing	3.3416	0.0208	2
previous	2.7797	0.0173	1
loan	1.8816	0.0117	1
default	1.5902	0.0099	2

Figure 6b: Variable Importance table (Importance and Relative Importance) from the same Data Exploration node.

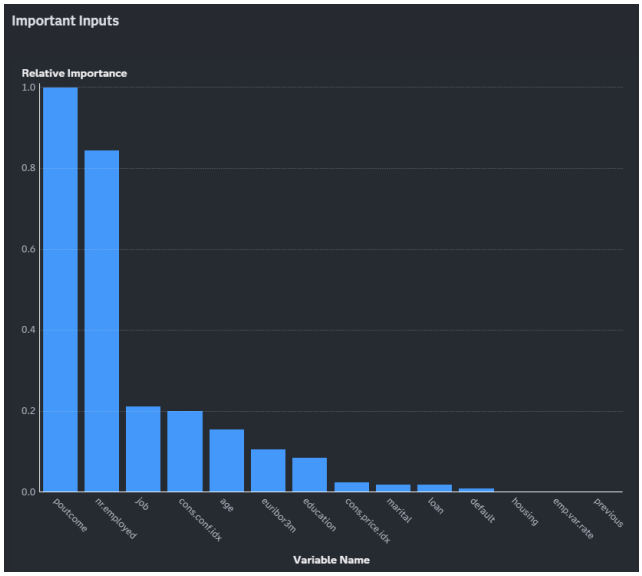


Figure 6c: Important Inputs chart after pdays is rejected (Data Exploration node after Manage Variables).

pdays has a skewness of −5.01 (Figure 2) because 96.4% of records equal 999 (Figure 5a; Table A1). The value 999 acts as a code for "not previously contacted" (Moro et al., 2014), so only 285 records hold real day counts. SAS ranks pdays as the most important input, with a relative importance of 1.0000 against 0.1029 for poutcome (Figures 6a and 6b). This ranking is not reliable here, for two reasons. First, it appears to come mainly from one split, recorded value versus 999. Among the 285 recorded values, subscription is 63%, 74%, 64% and 50% for 0–3, 4–6, 7–10 and 11 or more days, so the number of days shows no clear pattern (Table A1). Second, that split mostly repeats poutcome. All 253 customers with poutcome = success have a recorded pdays (Figure A2), and they subscribe at 67%, about the same as the 66% for all recorded-pdays customers (Table A1). Once pdays is rejected, importance shifts strongly towards poutcome (Figure 6c), consistent with substantial overlap between the two variables. pdays is also linked to previous, with a symmetric uncertainty of 0.2576 (Figure 5b).

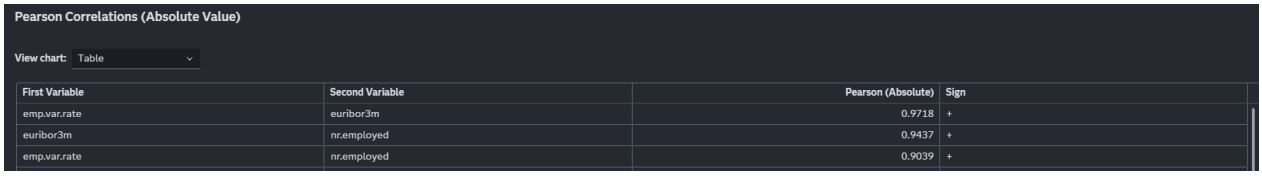


Figure 7: Correlation among the macroeconomic variables.

Table 1: Skewed variables, their importance and the decision taken

Variable	**Skewness (Figure 2)**	**Relative importance (Figure 6b)**	**What explains it**	**Decision**
pdays	−5.01	1.0000 (highest)	Driven by the 999 code; overlaps with poutcome	Reject
nr.employed	−1.07	0.7985 (second)	Mild skew; 11 distinct values; no outliers	Keep as is
cons.price.idx	2.16	0.0348 (low)	Separate tail of 160 values	Cap at 95.3725
previous	3.97	0.0173 (low)	Natural count; 86% zeros	No treatment

In line with the course approach, importance helps decide which skewed variables need action: a variable that is both highly skewed and important is investigated first (Table 1). nr.employed is also important, with a relative importance of 0.7985, but its skew is mild at −1.07 and it has only 11 distinct values (Table A1), so a transformation would add little. cons.price.idx and previous have low importance, at 0.0348 and 0.0173. cons.price.idx is still capped because its separate tail distorts the summary statistics, whereas previous, with a skewness of 3.97, is zero for 86% of customers, a natural count shape (Table A1). Three other points need no cleaning. The target is imbalanced, at 11.3% yes (Figure A3). emp.var.rate, euribor3m and nr.employed are strongly correlated, with absolute correlations from 0.90 to 0.97 (Figure 7). And “unknown” is a recorded answer, for example 21% of default (Table A1), so it is kept as its own category.

Table 2: Diagnosis summary and priority

Issue	Key evidence	Records affected	Priority	Decision
Missing age and education	~10% each	1,509 (18.9%)	High	Impute
cons.price.idx extremes	Skewness 2.16; 160 values above 95.3725	160 (2.0%)	High	Replace
pdays special code	96.4% equal 999; high importance explained by the code	7,715 (96.4%)	Moderate	Reject
Older ages (70+)	Plausible, gradual taper	84 (1.2% of observed)	None	No treatment
previous, nr.employed skew	3.97; −1.07	Natural structure	None	No treatment
Imbalance, strong correlation, "unknown" answers	11.3% yes; r up to 0.97	n/a	Deferred	Address in modelling

Table 2 summarizes each issue, its key evidence, its priority and the decision taken.

pdays is given the role Rejected, so it is left out of modelling, for the reasons given in Section 2. The alternatives fit worse: a log transformation would turn 999 into about 6.9 and keep the special code, capping would destroy its meaning, and a "recently contacted" flag would mark only 3.6% of records and overlap heavily with poutcome (Table A1). Rejection has a cost, as customers with a recorded value subscribe far more often, at 65.6% versus 9.3% (Table A1). Rejection therefore sacrifices some useful information, mainly the recorded-versus-999 distinction and exact recency, but avoids using a variable dominated by a special code.

Table 3: Resulting age distribution under alternative imputation values (calculated; see Table A1)

Method	Imputed value	Resulting mean	SD	Skewness
Observed (before)	n/a	40.06	10.45	0.78
Median (chosen)	38	39.85	9.93	0.87
Mean	40.06	40.06	9.91	0.82
Mode	31	39.15	10.28	0.93

Missing ages are filled with the median, 38. The alternatives are the mean of 40.06 and the mode of 31 (Figure 2; Table A1), and Table 3 compares the results. Ages are skewed towards older customers, which pulls the mean about two years above the median. The mean is also reasonable, so the choice rests on resistance to extreme values. The mode is less suitable because no age clearly dominates: the most common, 31, covers under 5% of observed records, and it pulls the mean down to 39.15.

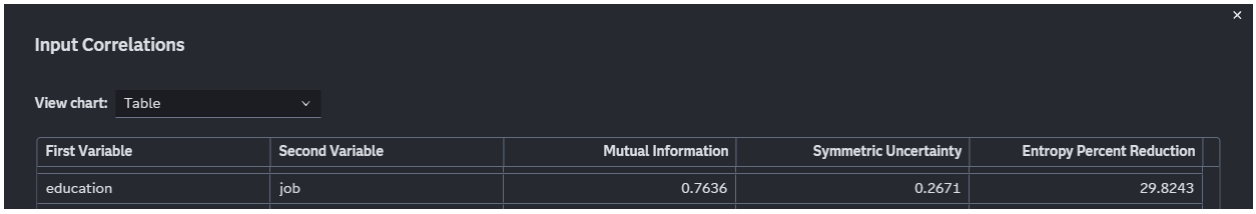


Figure 8: Association between education and job.

education is nominal with eight levels and about 10% missing. Filling with the mode would put all 800 missing records into university.degree, raising it from 29.2% of observed records to 36.3% of the dataset (Table A1). Tree imputation instead predicts each record's level from its other values (Ebersole, 2022). SAS finds that job is a useful clue to education. Its association scores, a mutual information of 0.7636, a symmetric uncertainty of 0.2671 and an entropy percent reduction of 29.82, all measure how much knowing one variable tells us about the other (Figure 8). Tree imputation is preferred because it can assign different levels to different types of customer.

cons.price.idx's 160 extreme values are replaced using an upper limit of 95.3725, the IQR fence, with no lower limit because no value falls below the lower fence. Deleting them would lose their other data. The IQR fence also sits inside the empty gap between 94.767 and 97.0, so it cleanly separates the group (Table A1). It is preferred because the extreme values do not distort it (Tukey, 1977).

No Transformations node is used. previous and nr.employed are skewed because of their natural shape, pdays is rejected, and cons.price.idx is rechecked after replacement. Strong correlation will be handled through model-specific variable selection, target imbalance through how models are fitted and assessed, and "unknown" answers are kept as categories.

4. Data Preparation Pipeline Design and Implementation

Variable settings were checked on the Data tab: y is the Target, categorical variables are nominal, apart from default, which is binary, and numeric variables are interval (Figure A6). The project uses a 60% training, 40% validation and 0% test partition, stratified by y (Figure A8). The Imputation node learns its filling rules from the training partition (Ebersole, 2022).



Figure 9: Data preparation pipeline in SAS Viya Model Studio.

Figure 9 shows the pipeline. After Explore Raw, Manage Variables sets pdays to Rejected, with a Data Exploration node confirming the change. Replacement then caps cons.price.idx, Imputation fills age and education, and Explore Post-Replace and Explore Final check the results.

The order matters for three reasons. First, role changes carry through to later steps, so once pdays is rejected it cannot affect Replacement, Imputation or the Tree used for education. Second, Replacement comes before Imputation so that imputation uses the treated REP_cons.price.idx. Third, Imputation comes last because it creates the IMP_ variables that replace the originals.

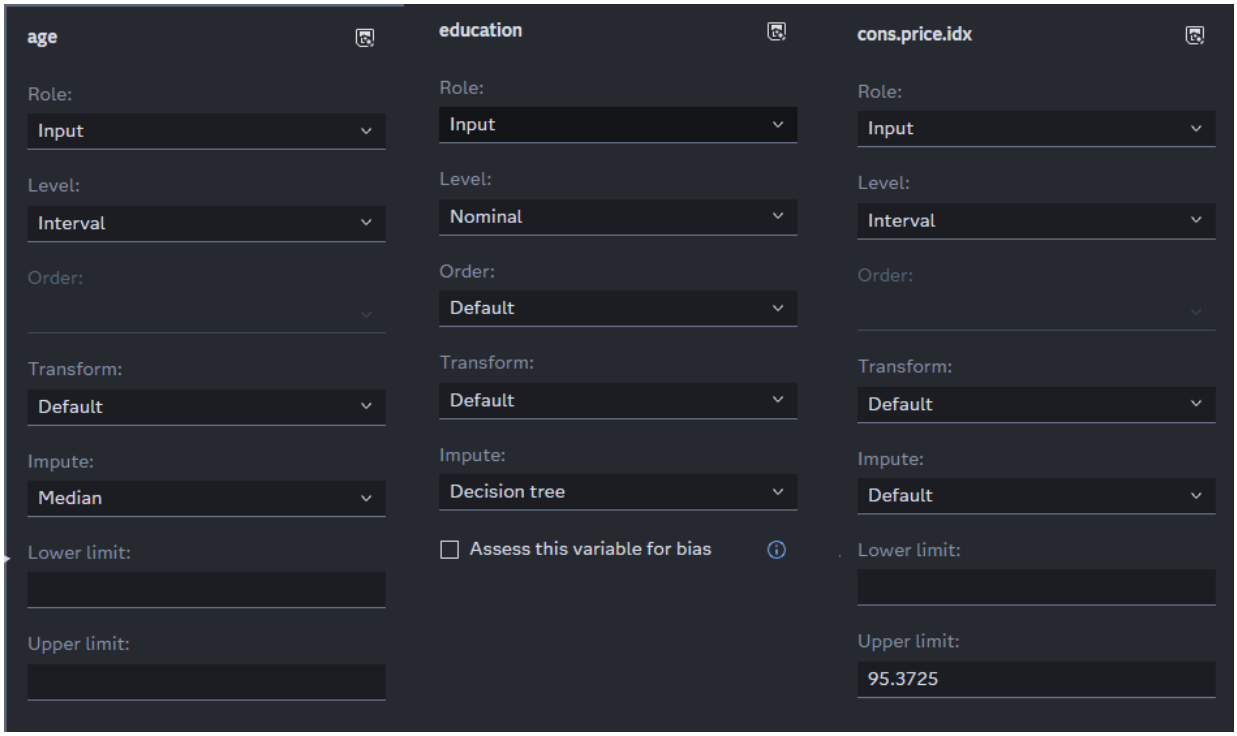


Figure 10a: Variable metadata settings for age (Impute: Median), education (Impute: Decision tree) and cons.price.idx (Upper limit: 95.3725).

Interval Variables

Name	Variable Label	Replace Variable	Limits Method	Lower Limit	Upper Limit	Replacement ...	Lower Replace...	Upper Replace...
cons.price.idx		REP_cons.price.idx	METALIMIT	.	95.3725	COMPUTED	.	95.3725

Figure 10b: Replacement node settings for cons.price.idx.

Imputed Variables Summary

Imputed Vari...	Method	Input Variable	Value	Numeric Value	Percent Miss...	Variable Level	Type	Variable Label
IMP_age	MEDIAN	age		38	10.0417	INTERVAL	N	Imputed age
IMP_education	TREE	education	<VARIES>	.	9.9167	NOMINAL	C	Imputed education

Figure 10c: Imputed Variables Summary.

Key settings were tuned in the variable metadata: age imputes with the Median, education with a Decision tree, and cons.price.idx has an upper limit of 95.3725 with no lower limit (Figure 10a). Replacement applies this limit, creating REP_cons.price.idx (Figure 10b). Imputation applies the Median, which is 38, and the Tree method, creating IMP_age and IMP_education (Figure 10c). Manage Variables sets pdays to Rejected (Figures A6 and A7).

5. Evaluation of Preprocessing Outcomes

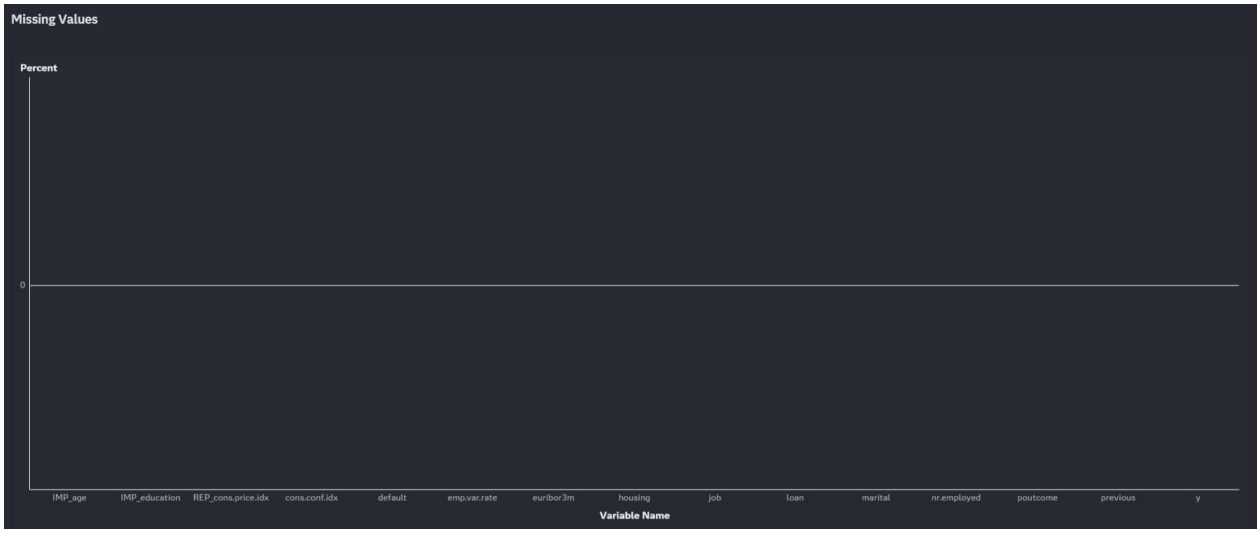
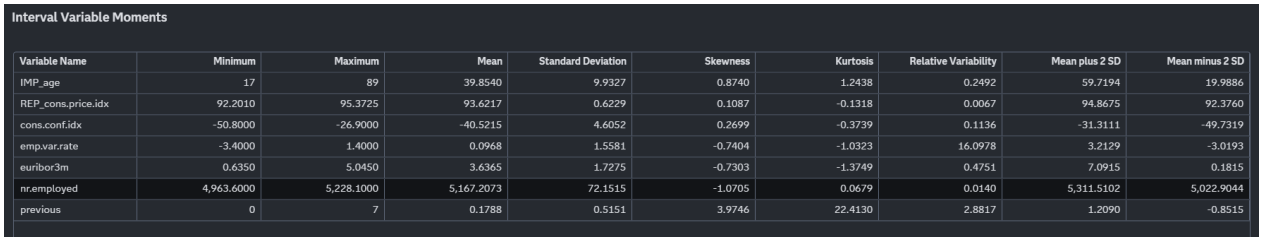


Figure 11: Missing values after preprocessing.

All active inputs now show 0% missing values (Figure 11), versus about 10% originally (Figure 1), so all 8,000 observations are complete. This also avoids the 18.9% record loss that complete-case deletion would cause.



Interval Variable Moments

Variable Name	Minimum	Maximum	Mean	Standard Deviation	Skewness	Kurtosis	Relative Variability	Mean plus 2 SD	Mean minus 2 SD
IMP_age	17	89	39.8540	9.9327	0.8740	1.2438	0.2492	59.7194	19.9886
REP_cons.price.idx	92.2010	95.3725	93.6217	0.6229	0.1087	-0.1318	0.0067	94.8675	92.3760
cons.conf.idx	-50.8000	-26.9000	-40.5215	4.6052	0.2699	-0.3739	0.1136	-31.3111	-49.7319
emp.var.rate	-3.4000	1.4000	0.0968	1.5581	-0.7404	-1.0323	16.0978	3.2129	-3.0193
euribor3m	0.6350	5.0450	3.6365	1.7275	-0.7303	-1.3749	0.4751	7.0915	0.1815
nr.employed	4,963.6000	5,228.1000	5,167.2073	72.1515	-1.0705	0.0679	0.0140	5,311.5102	5,022.9044
previous	0	7	0.1788	0.5151	3.9746	22.4130	2.8817	1.2090	-0.8515

Figure 12: Interval variable moments after preprocessing.

Table 4: Before and after statistics (SAS output)

Statistic	**age (before)**	**IMP_age**	**cons.price.idx (before)**	**REP_cons.price.idx**
Mean	40.0600	39.8540	93.6661	93.6217
Standard deviation	10.4498	9.9327	0.8012	0.6229
Skewness	0.7755	0.8740	2.1554	0.1087

Kurtosis	0.7856	1.2438	9.6095	−0.1318
Maximum	89	89	98.1261	95.3725

Median imputation changed the distribution of age only slightly: the mean moved from 40.06 to 39.85, the standard deviation fell from 10.45 to 9.93 and skewness rose from 0.78 to 0.87 (Table 4). Now 13.2% of records share the age 38, compared with 3.6% before (Table A1). One limitation remains: the filled ages ignore links with other attributes, since the observed mean age is 61.5 for retired customers and 26.4 for students (Table A1).

Table 5: Education level shares (%) and imputed counts (from Figures A4 and A5; mode column calculated, see Table A1)

Level	**Observed (excluding missing)**	**After Tree**	**Records imputed into level**	**If mode used (Table A1)**
university.degree	29.17	30.64	351	36.25
high.school	22.78	21.69	95	20.50
basic.9y	15.11	14.78	94	13.60
professional.course	12.63	12.91	124	11.36
basic.4y	10.64	11.15	126	9.58
basic.6y	5.51	4.96	0	4.96
unknown	4.15	3.86	10	3.74
illiterate	0.01	0.01	0	0.01

Tree imputation spread the 800 missing records across several education levels (Table 5; Figures A4 and A5). University.degree rose from 29.17% of observed records to 30.64%, compared with 36.25% if the mode had been used. Of the 800 filled cases, 351 went to university.degree, none to basic.6y and only 10 to unknown, so the filled values became somewhat more concentrated in university.degree.

The maximum for cons.price.idx fell from 98.126 to 95.3725, skewness from 2.16 to 0.11, kurtosis from 9.61 to −0.13 and the standard deviation from 0.801 to 0.623 (Table 4; Figure 12). The top-end problem is largely fixed, so no further transformation is needed. One limitation remains: the 160 capped records, 2.0% of the data, now share the single value 95.3725. The cap limits their influence but does not recover true values, and the near-zero skewness partly reflects this.

pdays no longer appears among the final variables, so its special code no longer distorts the data. The main cost is losing recency information, much of which poutcome partly carries. previous, with a skewness of 3.97, and nr.employed, at −1.07, are unchanged, showing that preprocessing was selective (Figure 12). These results show the targeted issues were addressed, not better prediction, which needs modelling in Assignment 3.

6. Final Dataset Summary

Table 6: Final prepared dataset (14 inputs and 1 target)

Variable	**Type**	**Role**	**Treatment applied**	**Final status**
IMP_age	Interval	Input	Median imputation (38)	Retained; replaces age
IMP_education	Nominal	Input	Tree imputation	Retained; replaces education
REP_cons.price.idx	Interval	Input	Upper-limit replacement (95.3725)	Retained; replaces cons.price.idx
job	Nominal	Input	None	Retained
marital	Nominal	Input	None	Retained
default	Binary	Input	None ("unknown" level kept)	Retained
housing	Nominal	Input	None	Retained
loan	Nominal	Input	None	Retained
previous	Interval (count)	Input	None	Retained
poutcome	Nominal	Input	None	Retained
emp.var.rate	Interval	Input	None	Retained
cons.conf.idx	Interval	Input	None	Retained
euribor3m	Interval	Input	None	Retained
nr.employed	Interval	Input	None	Retained
y	Binary	Target	None	Retained

Table 7: Variables excluded or superseded during preprocessing

Variable	**Treatment**	**Reason**
pdays	Role = Rejected (Manage Variables)	96.4% of records equal the special code 999, so its numeric coding is unsuitable; its high importance mostly reflects the recorded-versus-999 split, which poutcome partly carries
age, education, cons.price.idx	Replaced	Replaced by IMP_age, IMP_education and REP_cons.price.idx

7. Conclusion

The final dataset is ready for predictive modelling. All 8,000 observations have complete active inputs, the CPI extremes no longer distort the data, with skewness down from 2.16 to 0.11, and realistic observations such as older customers were kept without unnecessary transformation.

Important limitations remain. Filled values are estimates: median imputation puts 13.2% of records at age 38 and ignores links between age and other attributes, while Tree imputation gave few or no cases to small education levels. Capped CPI values are not true observations, and the CPI limit was set using all 8,000 rows, not the training partition alone. Rejecting pdays avoids a misleading special code but gives up recency information that may help prediction. The skewed previous variable, correlated economic indicators, and the 11.3% event rate should be considered when choosing and assessing models.

References

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AI Statement

I used Claude (https://claude.ai/) to cross-check my report structure against the marking rubric, draft and refine the wording of the report, check calculated values against the dataset, suggest suitable references, and format the Word document. I also used ChatGPT feedback (https://chatgpt.com/) to review the draft and point out weaknesses in my reasoning. I critically reviewed all suggestions, accepting only those that fit the dataset and assignment requirements, and rejecting or modifying others, for example the treatment of pdays. I ran the SAS Viya pipeline and took all screenshots myself, verified the reported results against my SAS output and the dataset, and checked that the final content was consistent with the assignment requirements and my own decisions.

Appendix

Table A1: How values not shown in screenshots were calculated from the CSV file

Value in the report	**Section**	**How it was calculated**
11.3% yes and 88.7% no	1, 2	Rows with y = yes (901) divided by 8,000.
91 missing both; 1,509 with a gap (18.9%)	2	Rows with both age and education blank: 91. Rows with either blank: 800 + 800 − 91 = 1,509. Then 1,509 ÷ 8,000.
Subscription of rows with a gap, about 11% to 12%	2	Share of y = yes among rows with a blank age (11.8%), a blank education (11.4%) or either (11.6%), compared with rows with no blanks (11.2%).
Gap of about 2.3 and the 160 extreme values	2, 3	Rows with cons.price.idx above 95.3725: 160 (2.0% of 8,000). The smallest of them is 97.05 and the largest value below the fence is 94.767, a gap of about 2.3.
84 customers aged 70 or over	2	Using the 7,200 observed ages, Q1 = 32 and Q3 = 47, so the fence is 47 + 1.5 × 15 = 69.5. Observed ages above 69.5: 84 (1.2%).
96.4% equal 999; 285 recorded; 3.6%	2, 3	Rows with pdays = 999: 7,715 (96.4%). Rows below 999: 285 (3.6%).
Subscription of 65.6% versus 9.3%	3	Share of y = yes among rows with pdays below 999 (65.6%) and among rows equal to 999 (9.3%).
Subscription by days since last contact	2	Among the 285 recorded rows, grouped as 0–3 days (94 rows), 4–6 (107), 7–10 (52) and 11 or more (32), the share of y = yes is 62.8%, 73.8%, 63.5% and 50.0%.
253 successes; 67% versus 66%	2	Rows with poutcome = success: 253, all with pdays below 999. Their share of y = yes is 67.2%, against 65.6% for all 285 recorded rows.

previous is 0 for 86%; 11 distinct values; 21% unknown	2	Rows with previous = 0: 6,900 ÷ 8,000. Distinct values of nr.employed: 11. Rows with default = unknown: 21.4%.
Median 38, mean 40.06, mode 31 (under 5%)	3	Median and mean of the 7,200 observed ages. The most common age, 31, covers 4.9% of them.
Share at age 38: 3.6% before, 13.2% after	5	257 of 7,200 observed ages equal 38 (3.6%). After 800 blanks are filled with 38: 1,057 of 8,000 (13.2%).
Mean age of retired and student customers	5	Average observed age of rows with job = retired (61.5) and job = student (26.4).
Table 3 (age alternatives)	3	Fill the 800 blank ages with 38, 40.06 or 31 in turn, then recompute the mean, standard deviation and skewness over 8,000 rows. The median row matches the SAS output for IMP_age in Table 4.
Education shares and the mode alternative	3, 5	Observed shares: level count ÷ 7,200 (for example, university.degree 2,100 ÷ 7,200 = 29.17%). Mode alternative: add the 800 blanks to university.degree and divide each count by 8,000 (2,900 ÷ 8,000 = 36.25%).

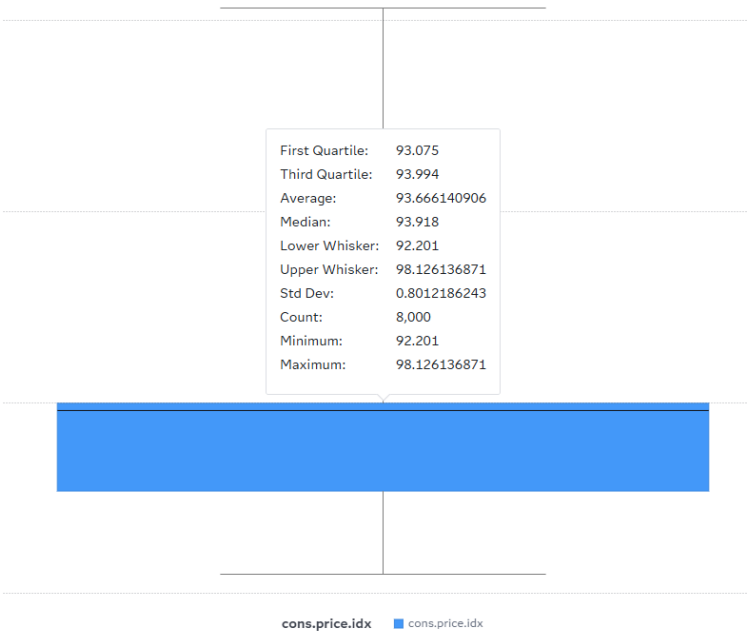
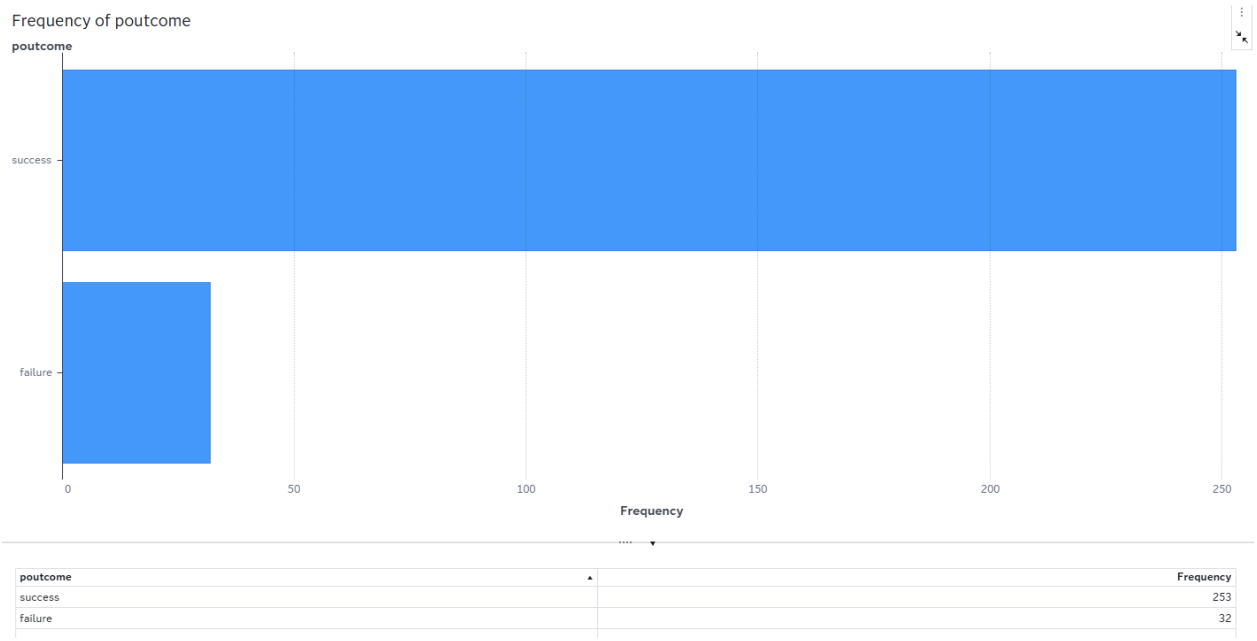


Figure A1: cons.price.idx box plot.



poutcome	Frequency
success	253
failure	32

Figure A2: Frequency of poutcome among records with pdays below 999.

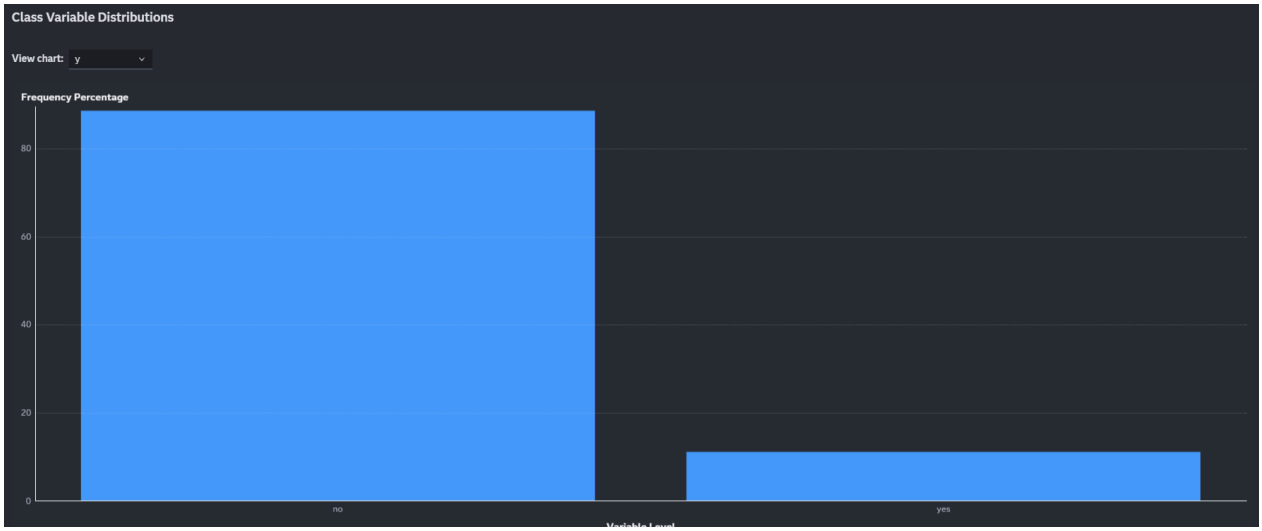


Figure A3: Distribution of the target variable y.

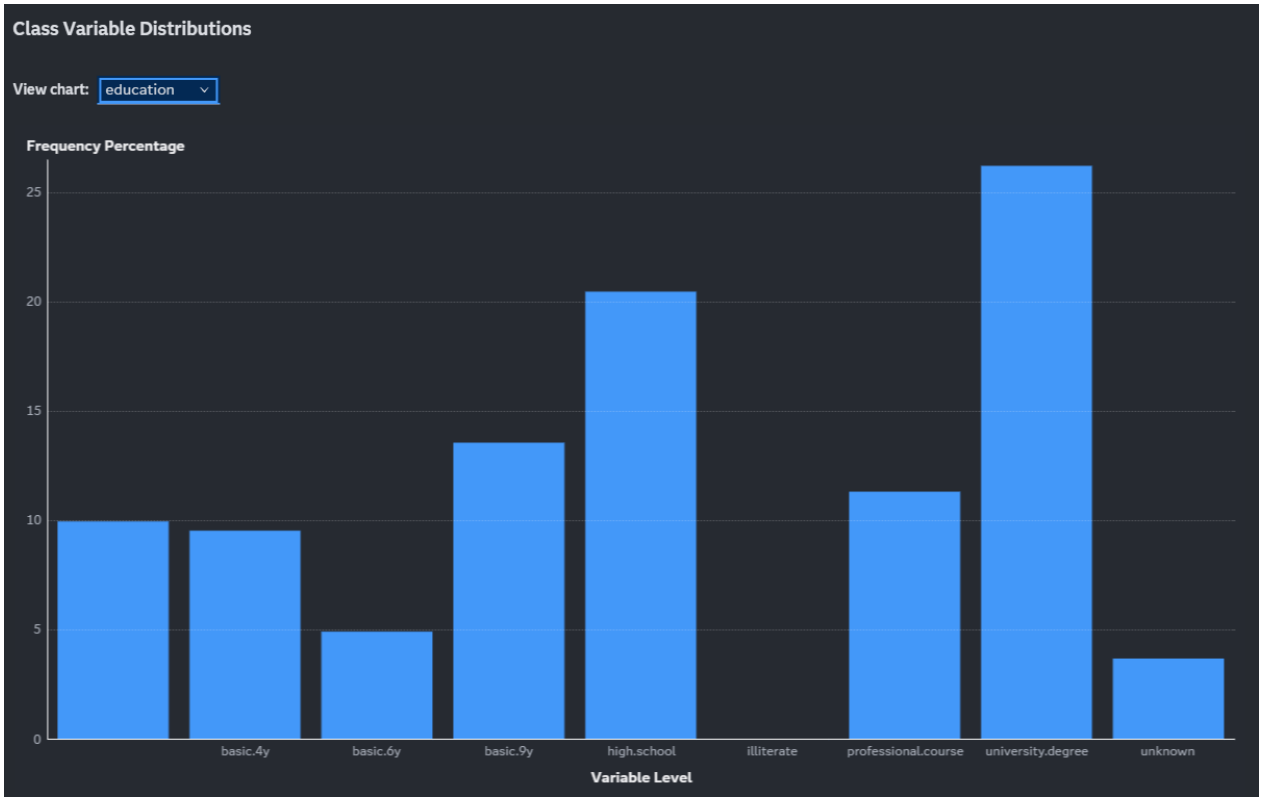


Figure A4: Education distribution before Tree imputation.

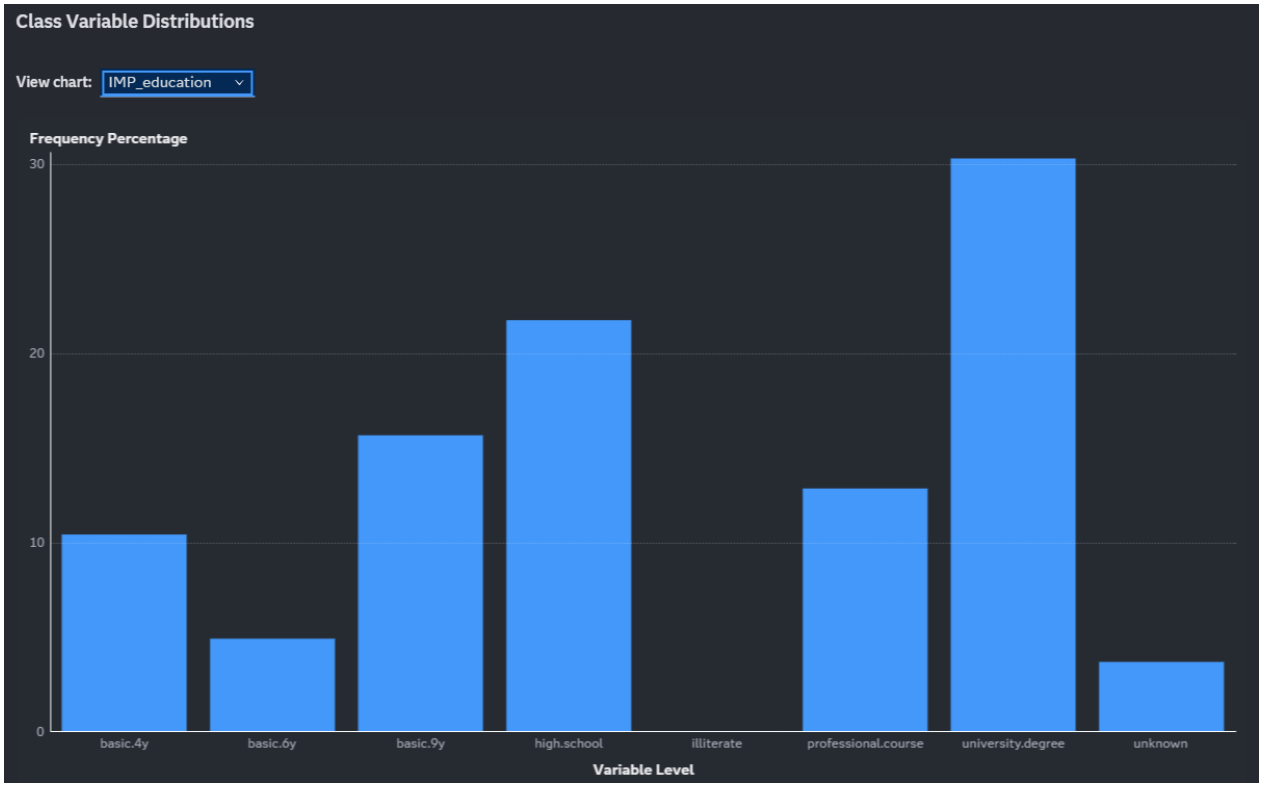


Figure A5: Education distribution after Tree imputation.

Variable Name ↑	Label	Type	Role	New Role	Assess for Bias	Level	New Level	Order
age		Numeric	Input			Interval		Default
cons.conf.idx		Numeric	Input			Interval		Default
cons.price.idx		Numeric	Input			Interval		Default
default		Character	Input			Binary		Default
education		Character	Input			Nominal		Default
emp.var.rate		Numeric	Input			Interval		Default
euribor3m		Numeric	Input			Interval		Default
housing		Character	Input			Nominal		Default
job		Character	Input			Nominal		Default
loan		Character	Input			Nominal		Default
marital		Character	Input			Nominal		Default
nr.employed		Numeric	Input			Interval		Default
pdays		Numeric	Input	Rejected		Interval		Default
poutcome		Character	Input			Nominal		Default
previous		Numeric	Input			Interval		Default
y		Character	Target			Binary		Default

Figure A6: Variable list showing roles and levels (y is the Target; pdays is assigned New Role = Rejected).

Metadata Changes

Name	Role	Level	Order	Transfo...	Impute	Lower li...	Upper li...	Uname	Constan...
pdays	REJECTED					.	.	PDAYS	

Figure A7: Manage Variables metadata changes (pdays set to Rejected).

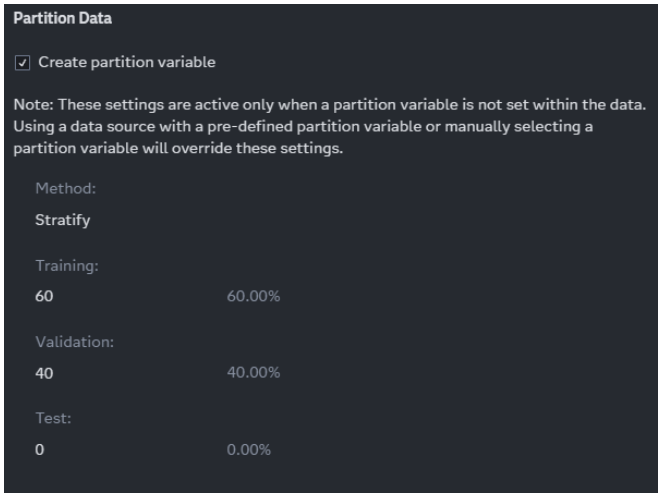


Figure A8: Partition settings (stratified; training 60%, validation 40%, test 0%).